

Project Design Phase Proposed Solution Template

Date	15 July 2026
Team ID	
Project Name	UCab Premium – Cab Booking Application
Maximum Marks	2 Marks

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Urban commuters struggle to book reliable, transparently-priced cabs, while drivers lack a structured system for receiving and managing ride requests — dispatch today is largely informal and manual.
2.	Idea / Solution description	UCab Premium is a MERN-stack cab booking web application with three role-based portals (Rider, Driver, Admin), JWT-secured authentication, a five-stage real-time booking lifecycle, transparent fare estimation, automated PDF receipts, and an admin analytics dashboard built with Recharts.
3.	Novelty / Uniqueness	A unified platform giving all three stakeholders — rider, driver, and admin — real-time visibility into the same booking lifecycle, combined with automated PDF receipt generation and built-in analytics, rather than fragmented manual or phone-based dispatch.
4.	Social Impact / Customer Satisfaction	Reduces uncertainty and anxiety for riders through live status tracking and transparent pricing; gives part-time/full-time drivers a structured, fair, and organized source of ride requests and income tracking.
5.	Business Model (Revenue Model)	Commission-based revenue on completed rides (a percentage of each fare), with potential future tiers for driver subscription plans or premium ride categories.
6.	Scalability of the Solution	The modular MVC backend, REST API (35 endpoints), and MongoDB data layer are designed to move from local hosting to cloud deployment (e.g., MongoDB Atlas), supporting growth in users, drivers, and ride volume without re-architecting the system.